

Lesson 2.5: Models and Solutions

Decision-making under uncertainty

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Models, solutions, values

	ω	$\bar{\omega}$
E	-1	-1
D	30	-10

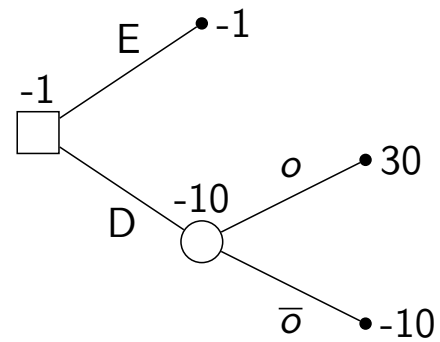
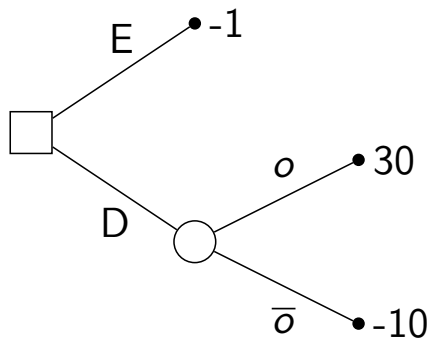
Decision rules:

- ▶ apply to models
- ▶ produce *solutions*

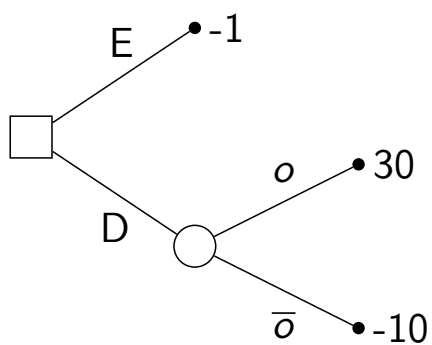
Example (Maximin)

- ▶ solution is E
- ▶ value is -1

Evaluating decision models

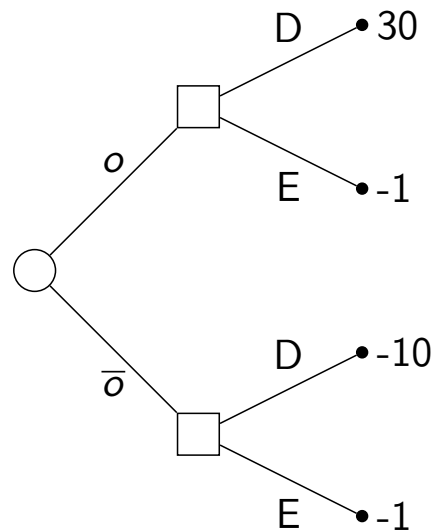


Information



Alternatives:

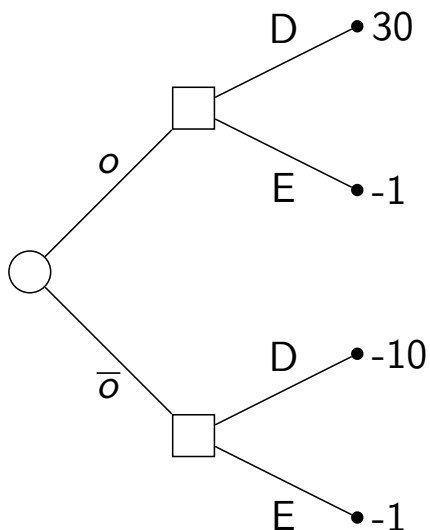
- ▶ E
- ▶ D



Additional alternatives:

- ▶ if o, then E; otherwise D
- ▶ if o, then D; otherwise E

Contingency plans



	o	$\bar{o}$
E	-1	-1
D	30	-10
if o , then E ; otherwise D	?	?
if o , then D ; otherwise E	?	?

Definition (Contingent action)

A *contingent action* or *conditional action* is an action that combines other actions depending on a contingency or condition. A contingent action is also called a *strategy* or *plan*.

Information sets

